

CS 232

Computer and Communication Networks

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Instructor:

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Office hours: by appointment (preference
1:30-3:30pm on Thursday)

I am italian!!!



Did he just say “spaghetti” or “network configuration”????

TA:

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Office hours: -:-

Networks:

- Architecture
 - Applications
 - Protocols
 - LANS
-
- No: Physical layer, hardware, coding, etc.

Objectives:

- Notions
- Modeling/understanding
- Design principles

Pre-requisites

- CS 132, EE 148
- Or similar
- Probability theory

Revise: probability, distribution
(exponential distribution),...

Take notes:

- You won't find ALL the material in the book/slides!!!!
 - E.g., problems
- I won't ask you “numbers”, but understanding and knowledge of the material

Keep up with readings

- You'll retain more information from class
- You'll be able to ask questions and clarify doubts
- You'll make me happy!

Website

Piazza - I'll send you an invitation

- Additional material
- Slides, outlines, etc.
- Discussion

Evaluation

- Assignments (problems) 40%
- Midterm (questions) 20%
- Final (questions) 40%

Comprehensive exam:

The CS232 portion of the CS MS Comprehensive Exam will consist of the course final. Students scoring a B+ or better will pass the CS 232 portion of the comprehensive exam.

You must let me know by October 15 if you wish to attempt the CS 232 portion of the CS MS Comprehensive Exam

Send an email to htang6@uci.edu and fangqil2@uci.edu

Networks

- Architecture
- Applications
- Protocols
- LANS

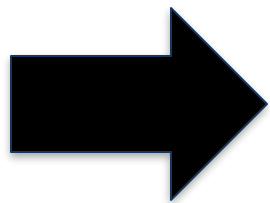
Architecture

Part 1:

Telephone network and circuit switching

Telephone networks

- 1 application (voice)
- Long unicast connections/sessions (calls)
- Almost constant traffic generation during the connection
- Sparse call arrival



Network design

Topologies

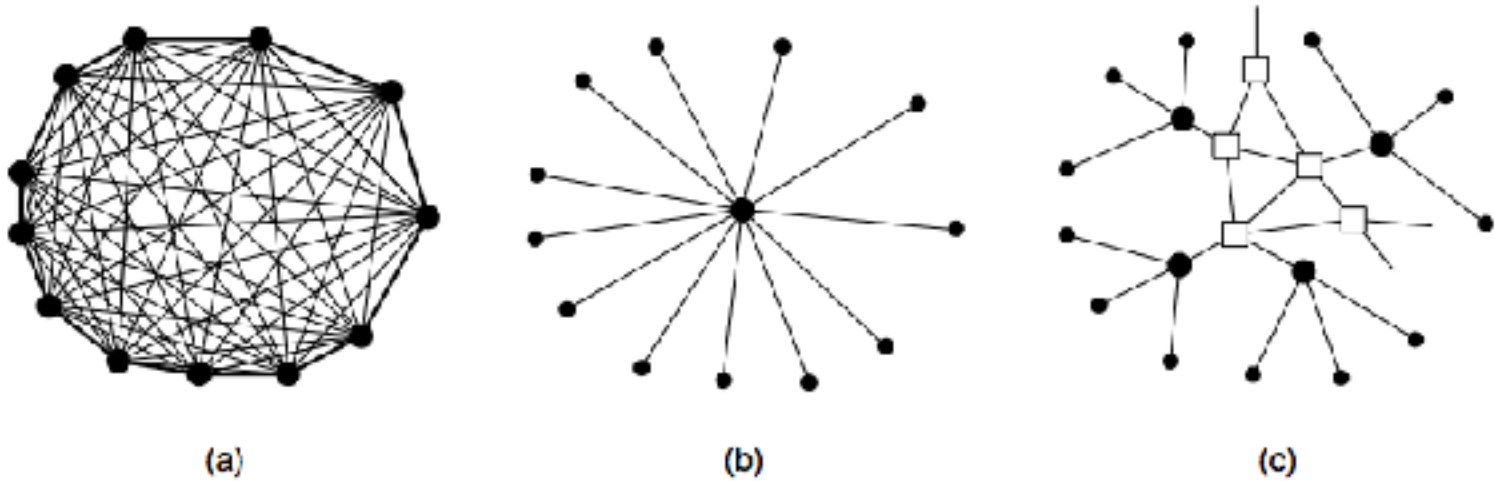
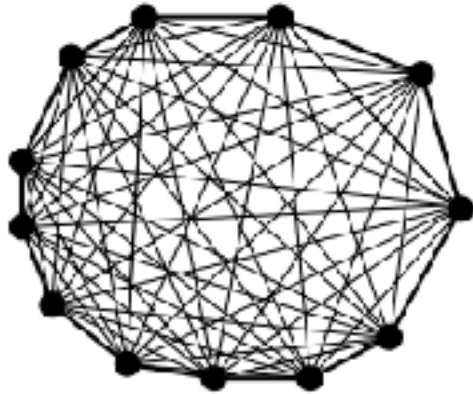


Figure 2-29. (a) Fully interconnected network. (b) Centralized switch.
(c) Two-level hierarchy.

Tanenbaum fig. 2-29

Fully-connected topology

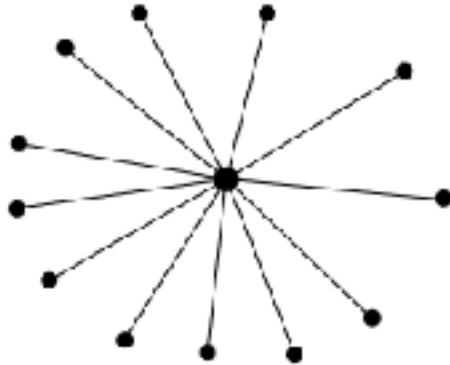


- Large number of links



Expensive infrastructure!

Centralized switch topology

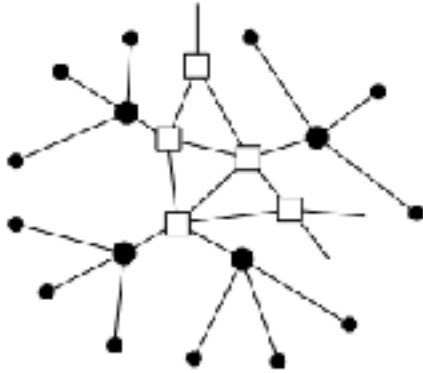


- Small number of links

Local infrastructure:

- Connections from any point to a central switch
- Very long wires

Multi-level hierarchical topology



- Slightly larger number of links (2 levels)

Scalable infrastructure:

- Many short connections
- Fewer long-distance connections

Topology with long distance

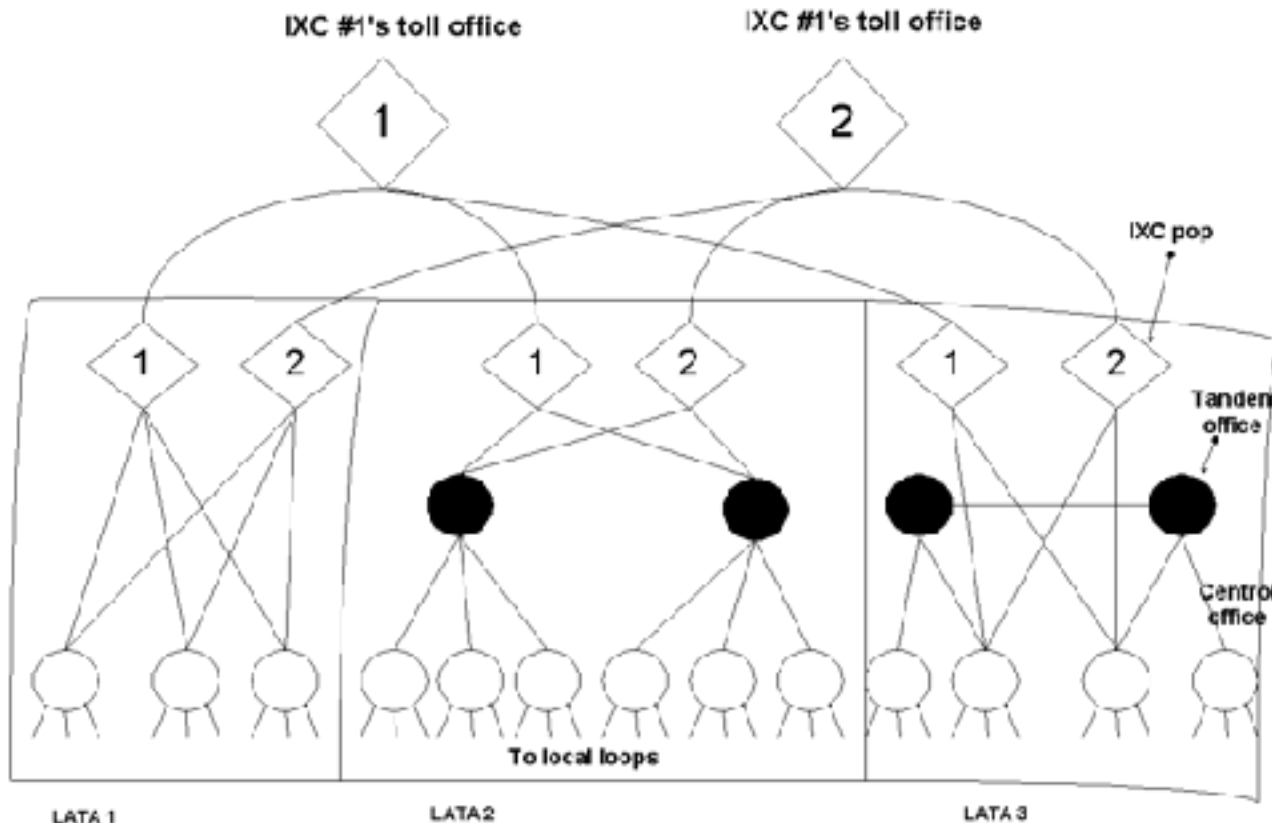


Fig. 2-16. The relationship of LATAs, LECs, and IXCs. All the circles are LEC switching offices. Each diamond belongs to the IXC whose number is in it

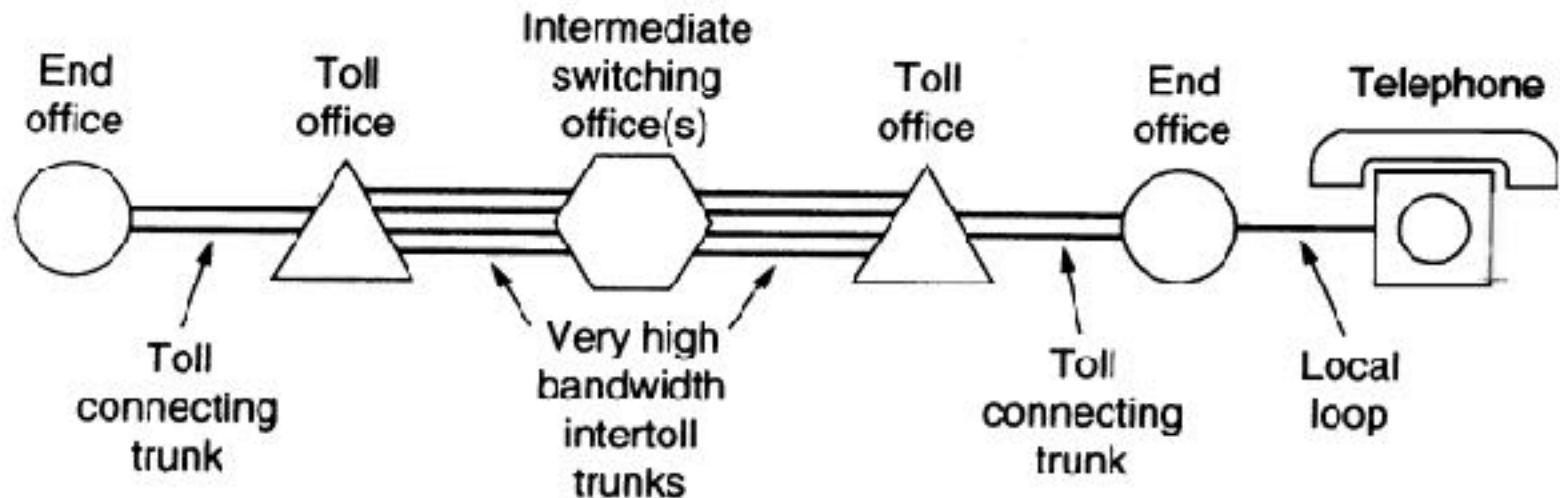
Multiple long-distance carriers

LATA = Local Access and Transport Area

LEC = local exchange carrier

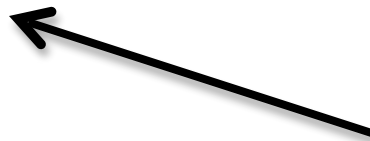
IXC = Interexchange (long distance) carrier

Lines and trunks



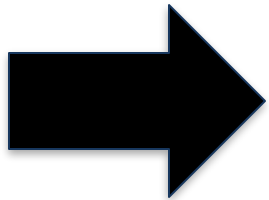
Tanenbaum fig. 2-15

Local loop line is often twisted-pair copper (low capacity, analog)
Trunks are often fiber (high capacity, digital)

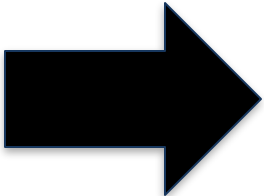


Trunks

- Trunk lines carry more than 1 call simultaneously (thousands-millions)

 **High capacity**

- Digital vs Analog

Long distance  **Signal degrades with distance**

- Analog sensitive to noise
- Digital: regeneration, recovery